National University of Computer and Emerging Sciences 0024

Islamabad Campus

Numerical Computing (CS2008)

Course Instructor(s):
Dr. Muhammad Rizwan
Section(s): BS(CS) A 2022

Sessional-I Exam

Total Time (Hrs): 1
Total Marks: 50
Total Questions: 5

Date: Sep 23, 2025

22i-2308 — **Roll No**

A — **Course Section**

Student Signature

Do not write below this line.

Attempt all the questions.

CLO 3: [Demonstrate the capability of using numerical analysis library...]

Q1: **[2+8=10 marks]**

a.) Matrix A created via np.array([[32,64,86],[86,64,95],[55,55,66]]).

$$A = \begin{pmatrix} 32 & 64 & 86 \\ 86 & 64 & 95 \\ 55 & 55 & 66 \end{pmatrix}$$

What do the following two slices extract from the array? Write portion of A.

1. A[0:2, 2:3]
2. A[0:2, 1:2]

1.	2.
$A_{01} : A_{02}$ ✗	$A_{22} : A_{23}$ ✓

0

b.) Complete the table by writing the appropriate SciPy.Optimize functions names of the following mentioned numerical methods. It is required to fill in the table below and mention the core mathematical concept behind the method in simple mathematical or descriptive format.

Function	Parameters /Arguments	SciPy.Optimize function name	Output	Core Concept
Bisection	int a, int b	bisect	int c	uses bisection method to give midpoint
Fixed-point	int a, int b	fixedpoint	int c	uses fixed points to calculate close answer
Newtons'	int a	newtons	int c	uses newtons method to find root
Secant	int a, int b	secant	int c	uses secant method to find root

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[CLO 2: Analyze Sources of Error and behavior of different operators of numerical methods...]

Q2: (a) List and briefly explain the different types of errors that occur in numerical computation (with example) **[4.5+2.5+3=10 marks]**

Round off	occurs when two numbers operation does not yeild an acurate enough result. 0.1+0.22 0.30 1
Truncation	occurs when numbers cannot be accurately converted to binary 1/10 = 1/2.5 Can't be further divided
computer Epsilon	occurs when the difference between non zero values in a number is great For example, 1.000 1

3.5

(b): If the relative error in measuring an area of 350m was 0.04, calculate the absolute error for that measurement.

1Km zero 0.5

(c): Calculate the truncation error in calculating f'(2) for $f(x)=x^2$ by $f'(x) \approx \frac{f(x+h)-f(x)}{h}$ with h=0.2.

zero 1/5

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[CLO 1: Apply numerical methods for various mathematical operations…]

Q3: **[10 marks]**

Using the Bisection Method, perform three iterations to approximate the root of the following function within the interval [a,b]:

$$x^3 + x^2 - 1 = 0$$

with the initial interval [a,b]=[0, 2]
For each iteration, show the values of a, b, c=a+b/2, and f(c).
Finally, after three iterations, specify the new interval where the root of the function lies.

Iter#	a	b	c	f(a)	f(b)	f(c)
1	0	2	1	-1	11	1
2	0	1	1/2=.5	-1	1	-.625
3	.5	1	.687	-.625	1	-.203
4	.687	1	.843	-.203	1	.3097

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[CLO 1: Apply numerical methods for various mathematical operations…]

Q4: Use Newton's method $x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$ to find a solution of the following, accurate to within 10^{-2}

$2x + 3\cos x - e^x = 0$, for $1 \le x \le 2$, by taking $x_0 = 1.5$ **[10 marks]**

$x_0 = 1.5$

$f'(x_k) = 2 + 3\sin x - e^x$

$x_1 = 1.5 - \frac{1.517}{2.156}$

$= 1.5 - 0.7036$

$= 0.7963$

Hence, $x \approx 1.925$

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[CLO 1: Apply numerical methods for various mathematical operations]

Q5: Write a Python program to implement the Secant Method $x_{k+1} = x_k - \frac{f(x_k)(x_k - x_{k-1})}{f(x_k) - f(x_{k-1})}$ **[7+3 marks]**

Define a function name my_secant that performs the root-finding procedure. Additionally, provide a separate code segment to test your implemented function

Code area for my_secant function to solve roots using Secant Method

```
mysecant (int x1, int x2)
{

}
```

Write test code, which calls your method to solve the equation mentioned in Q4, with $x_0=1$, $x_1=1.6$, and $\varepsilon=10^{-2}$

$x_0 = 1$, $x_1 = 1.6$, $\varepsilon = 10^{-2}$

$2x + 3\cos x - e^x = 0$ where $1 \le x \le 2$

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